A Trait experiment

200 seeds per primary group
4 prespecified arms

Search-pipeline and mutation comparisons

P1: mean population
P2: terminal extinction

B Neural experiment

10 runs per method
5 core methods + ES-fixed addendum

OE - PPO-DR adaptation IQM:
-0.559 [-0.609, -0.520]

Zero-shot and adaptation tests

C Inference scope

Separately tuned pipelines
One world and variation schedule

Matched nominal agent steps
Whole-run uncertainty

Historical reconstruction:
separate supporting evidence